

Hermes 4 Technical Report — Digest for Our Build

Nous Research's Aug 2025 post-training report — takeaway: every quality problem they solved, they solved in code (schemas, token caps, template tokens), not in prompts.

⚡ TL;DR — 5 things to remember

1. **The 14B hit its 40,960-token context 60% of the time** on LiveCodeBench — small models overthink until they return nothing, and the fix was a forced stop token, not "be concise" (§3.1).
2. **The judge never shares weights with the generator** — self-favoring judges are a known failure mode, with the paper cited (§2.1.2, [41]).
3. **Parsers lie more than models do**: 7.3% disagreement between a popular framework's GPQA parser and GPT-4o grading (§4.3.1).
4. **Errors halt the run, they never score as wrong** — a timeout shows up as a bug, not a fake quality drop (§4.3.1).
5. **Swapping the template's assistant role token for me** produced a first-person persona with fewer disclaimers — voice is structural, not prompted (§5.3.2).

📖 The Report in 60 Seconds

Hermes 4 is a family of open-weight reasoning models (14B / 70B / 405B) post-trained on Llama. The report covers **how the data was made and how it was measured**, not architecture. DataForge is a graph pipeline where each node declares PDDL pre/postconditions, so the graph wires its own edges; it produced ~5M samples / 19B tokens, reasoning samples averaging 5× more tokens each. Rejection sampling ran against ~1,000 task-specific verifiers inside Atropos, their async RL-environment controller. Training masks loss to assistant tokens only, and a separate fine-tune taught the 14B to stop reasoning at ~30k tokens by unmasking exactly two tokens. Evaluation runs through one shared OpenAI-compatible endpoint using single-file scripts. A final section documents behavior: persona fidelity, anti-sycophancy, template-level steering.

🔑 Key Learnings (by section)

Data (§2) - Nodes declare PDDL pre/postconditions and the DAG assembles itself; single-in/single-out means a graph is also a node (§2.1, §2.1.3). - **Cheap filter first**: ModernBert dedup at **cosine 0.7**, then an LLM judge (§2.1.1). - **Generator ≠ judge, always** — and they **train on the scaffolding**, keeping all intermediate LLM calls (§2.1.2). - **Format is scored separately from correctness**; the schema env **compiles Pydantic models on the fly**, scoring 1.0 only if output instantiates (§2.2.1, §2.2.4). - **Tool calls live inside a single <think> turn** — multiple round-trips, one assistant message (§2.2.5).

Training (§3) - Only assistant-role tokens get gradient (§3). - **The fix teaches exactly one token**: force `</think>` at 30,000 tokens, unmask only `</think>` and `<eos>` — the reasoning text gets zero gradient (§3.1). - **Truncation trades accuracy for reliability** (Table 2): AIME'25 -12.7%, overlong collapsed (AIME'24 28.2→6.1), **LCBv6 rose +54.5%** (28.6→44.2) (§3.1.3). - **70B and 405B needed none of it** — a small-model pathology (§3.1).

Evaluation (§4) - A score measures the model AND the engine — one shared endpoint, no harness bundling its own (§4.1). - **Log the extracted span, its score, and rejected candidates** — the 7.3% gap means "regression" is often your regex (§4.3.1). - **Single-file evals beat a framework**; duplication is the accepted price of readability (§4.3.1). - **Sampling settings are part of the result** (temp 0.6 / top-p 0.95 / top-k 20), and **RefusalBench** — 166 hand-written prompts, Sonnet 4 judge — makes refusal measurable (§4.5, §4.5.1).

Behavior (§5) - Policy rigidity is the default failure mode — frontier models re-inject AI-identity disclaimers on clearly fictional prompts; Hermes stayed in character (§5.1). - **Anti-sycophancy changed the reasoning, not just the manners** (§5.3.1). - **Style transfer reaches rhythm and diction**, with near-verbatim paraphrase flagged as the failure

mode (§5.2). - **Plasticity doesn't show in standard benchmarks** — needs qualitative probing (§5.4). *Caveat*: §5.1 says "five genres", names four.

Ecosystem (§6 + refs) - Compute is sponsored stage by stage: Chutes (rejection-sampling inference), Modal (LCB evals), Morph Labs (terminal-agent data). - **The stack is other people's maintained tools** — Axolotl [2], LightEval [11], vLLM [20], pydantic-cli [32]. **Atropos [27]** is the one thing they built. - **Third-party verification is a requirement** — Sam Paech independently re-ran EQBench3 and the writing benchmarks. - **Verifiability is the through-line**: Lean 4 [4], PDDL [9], IFEval [62] and successor [43] — rewards a program can check.

What It Means for Our Build

Agent runtime (gateway, crons) - Reasoning-token ceiling in code, not the prompt — a cron that silently burns context returns nothing *and* charges full price, ~99 times a day (§3.1). - **On cap-hit, force a stop marker and ask for the answer** — salvage at 1× cost, not 2× (§3.1.1). - **Timeouts surface as errors, never score 0** — otherwise DeepSeek flakiness reads as quality decay and pushes spend to Opus 5 (§4.3.1). - **Allow multi-tool round-trips in one assistant turn** — turn-per-call fights the trained distribution (§2.2.5).

Prompts & personas - Anti-sycophancy gets a real system-prompt block — on Hermes it reaches the reasoning, so the health coach actually disagrees (§5.3.1). - **Audit persona prompts for disclaimer-inviting language** ("as an AI") — rigidity is largely prompt-induced (§5.1). - **Character breaks? Suspect the role token** — the `assistant` → `me` swap is structural, where the gateway owns the template (§5.3.2). - **Route persona surfaces (Telegram bot, coach) to Hermes, keep Opus 5 for review** — contextual fidelity is the mechanism (§5.1).

Evals & quality - Pin the endpoint, not the library — so "DeepSeek is worse here" is the model, not two client stacks (§4.1). - **Log the extracted span on every graded run** — a 7.3% parser gap dwarfs most real regressions, and Concierge/Toto output is parsed downstream (§4.3.1). - **Track `overlong@limit` as its own axis** — score alone hides "never answered" (§3.1.3). - **House benchmark on the RefusalBench pattern** — ~150 hand-written prompts, LLM-judged; tone drift is invisible to algorithmic verifiers (§4.5.1).

Data & memory - Cosine-0.7 dedup ahead of any LLM pass in spine memory and the Toto/marketing corpora — a straight cost lever (§2.1.1). - **Never let a model judge its own output** — generate on DeepSeek flash, judge on Opus 5, cross-model always (§2.1.2, [41]). - **Schema-validate structured outputs, binary pass/fail** — "does it instantiate" is code-level enforcement (§2.2.4). - **Pre/postconditions on skill manifests** let the gateway *compute* legal chains instead of hand-maintained routing (§2.1, §2.1.3).

Content & products - Feed Substack/LinkedIn 2–3 style exemplars, not a topic brief — transfer reaches rhythm, which a brief can't specify (§5.2). - **Near-verbatim-overlap check before publishing** — §5.2 names paraphrase-of-source as a real observed failure. - **Persona set to stress-test Marketing Concierge** — synthetic users send messy traffic before real ones do (§2.3.2).

Ecosystem & tooling - Atropos [27] — same lab as Hermes Agent; the native layer if crons should emit RL-gradable trajectories (§4.3). - **LightEval [11]** — HF-maintained, light enough for the M1; replaces eyeballing with a regression gate. - **IFEval [62] + [43]** — checkable instruction compliance for Telegram output: machine-parseable = machine-checkable. - **Modal / Chutes burst compute** — grade on Zo while the M1 keeps generating (§4.3.3).

Consolidated Action Items

#	PRIORITY	ACTION	BENEFIT	WHY (SECTION)
1	P0	Max-reasoning-token cap per cron, in gateway code	Kills silent context burns	60% overlong, invisible until measured (§3.1)
2	P0	Log <code>overlong@limit</code> beside success rate	Separates "wrong" from "never answered"	Independent axes in Table 2 (§3.1.3)

#	PRIORITY	ACTION	BENEFIT	WHY (SECTION)
3	P0	Surface timeouts as errors, never score 0	Stops phantom drops pushing spend to Opus 5	Explicit error semantics (§4.3.1)
4	P0	Enforce generator $\neq$ judge in every eval/review cron	Kills self-favoring bias	Known failure mode (§2.1.2, [41])
5	P0	Schema-validate structured outputs, binary pass/fail	Parse failures become caught errors	Reward requires instantiation (§2.2.4)
6	P0	Route evals through one endpoint; stamp model + params + date	Comparisons stay valid	Score depends on engine + config (§4.1, §4.5)
7	P0	Log extracted span + rejected candidates when grading	Parser bugs $\neq$ model regressions	7.3% parser/judge gap (§4.3.1)
8	P0	Anti-sycophancy block in coach + PA system prompts	Agents that actually disagree	Changes reasoning, not tone (§5.3.1)
9	P0	Audit persona prompts for disclaimer-inviting language	Fewer character breaks	Rigidity is prompt-induced (§5.1)
10	P1	On cap-hit, force stop marker and request the answer	Salvage at $\sim 1\times$ cost, not $2\times$	Answers fine from partial traces (§3.1.1)
11	P1	~ 150 -prompt house benchmark, Concierge + health coach	Catches tone/refusal drift	RefusalBench pattern (§4.5.1)
12	P1	Cosine-0.7 dedup before LLM passes in spine memory	Cuts spend on near-duplicates	Cheap filter first (§2.1.1)
13	P1	Pre/postconditions on skill manifests, derive chaining	Composition without hand-wired routing	Declarative DAG edges (§2.1)
14	P1	Test <code>assistant</code> $\rightarrow$ <code>me</code> template swap on one persona	First-person voice, fewer disclaimers	High token sensitivity (§5.3.2)
15	P1	LightEval on 20 golden prompts, grading on Zo	Regression detection, wall-clock down	Keep runs inference-bound (§4.3.3, [11])

Sources

- The report: https://nousresearch.com/wp-content/uploads/2025/08/Hermes_4_Technical_Report.pdf
- Chunk digests: `out/chunk-1.md` ... `out/chunk-5.md` (same directory)